TEAM HANDBOOK

Ratiio

Everything you need to know about what we are building, how it works, and why it matters.

Product	Ratiio — ASI Voice-First AI Platform
Core AI	ARIA Architecture on Qwen 2.5 7B (fine-tuned)
Interface	100% Voice — No keyboard, no screen input required
Servers	Own hardware at CtrlS Hyderabad + Cloud burst (RunPod)
Document	Full product, tech, business, and team guide

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1 What is Ratiio?

The One-Line Answer

Ratiio is a voice-first AI platform powered by our own fine-tuned large language model — ARIA — that lets people control their entire digital and physical life using only their voice, with no keyboard, no touchscreen, and no screen input of any kind.

The Problem We Are Solving

Every AI assistant that exists today is fundamentally a text product with a voice layer painted on top. You speak, it transcribes, it responds in text, you read. That is not a voice-first experience — that is a keyboard replacement.

The real problem is deeper. Existing AI assistants are:

- Generic — they know nothing about you across sessions. Every conversation starts from zero.
- Reactive — they wait to be asked. They never tell you what you need to know before you know you need it.
- Unsafe — they can be manipulated by malicious content in emails, web pages, or clever phrasing to act against your interests.
- Fragmented — you use five different apps to do what one intelligent voice layer should handle entirely.
- Expensive for what they deliver — you pay for a generic product and get a generic product.

The Ratiio Solution

Ratiio is built from the ground up as a voice-first intelligence. Not a text product with voice on top. Not a wrapper around someone else's API. Our own model, our own servers, our own product — designed so that speaking to Ratiio feels like talking to the most capable, most trusted person you have ever worked with.

> **The Ratiio Promise**
>
> If you can think it, you can say it. If you can say it, Ratiio does it. No typing. No tapping. No navigating menus. Your voice is the only interface you will ever need.

Why Now?

Three things have converged to make Ratiio possible in 2024 that were not possible before:

Factor	What Changed	Why It Matters for Ratiio
Open-source LLMs	Qwen 2.5, Llama, Mistral — powerful models available to anyone	We can fine-tune a world-class 7B model without billion-dollar compute budgets
Efficient fine-tuning	QLoRA and LoRA techniques allow fine-tuning on consumer-grade GPUs	We train ARIA on Kaggle for free — no cloud bill needed during development
Voice technology maturity	Whisper (STT), ElevenLabs (TTS), Silero VAD are production-ready and affordable	The voice pipeline we need costs almost nothing at launch scale
India's AI moment	350 million smartphone users, growing AI awareness, low cost of living for talent	We can build world-class product with India-based team and serve a massive market
Subscription model viability	Indian users now pay for SaaS — Spotify, Netflix, Zepto Pro — habit is formed	Rs 999/month subscriptions are realistic and sustainable

Who Is Ratiio For?

User Type	How They Use Ratiio	Their Pain Today
Working professional	Voice-commands emails, calendar, tasks, research while commuting or cooking	Switches between 8 apps, types constantly, loses context between tools
Student	Research, note-taking, study planning, language learning — all by voice	Generic AI gives generic answers with no memory of their learning journey
Entrepreneur	Manages comms, planning, decisions, team coordination by voice	No AI that understands their business context and remembers their decisions
Person with disability	Full digital life access without physical keyboard or screen interaction	Current voice assistants are too limited for real productivity
Tech team member	Code generation, debugging, documentation, API testing by voice	Context loss between sessions means re-explaining the same project repeatedly

2 The Name — Ratiio and ARIA

Two Names, One Product

Every team member needs to understand the distinction between these two names clearly, because they will come up constantly in your work.

	Ratiio	ARIA
What it is	The product — the brand — what users and the world see	The internal technical architecture — what engineers build and reference
Full meaning	Our startup name and product identity	Adaptive Reasoning Intelligence Architecture
Who uses it	Users, investors, press, marketing, everyone external	Engineering team, training scripts, system prompts, internal docs
Lives where	App, website, pitch deck, marketing, AppStore	Codebase, model configs, system prompts, training blueprints
When to use it	Anytime you talk about the product publicly or to users	Anytime you talk about the model internally — training, prompts, configs

The Analogy

Think of how BMW works. BMW sells cars to customers — that is the brand people know. Under the hood, engineers build on a platform called CLAR. Customers never hear about CLAR. Engineers never lose sight of it.

Ratiio is BMW. ARIA is CLAR. Users experience Ratiio. Engineers build ARIA.

> **What Ratiio Says When Asked Its Name**
>
> When a user asks 'what is your name?' — Ratiio answers: I am Ratiio. That is the end of the answer. Users do not need to know about ARIA, Qwen, fine-tuning, or any of the internal stack. The product is Ratiio. The intelligence inside is ARIA. Only the team needs to know both.

Naming Rules for the Team

- In user-facing copy, marketing, and the app UI: always say Ratiio
- In system prompts and model configs: 'Your name is Ratiio. You are powered by ARIA architecture.'
- In training scripts, code, and internal docs: ARIA is fine to use
- In investor decks: Ratiio as product, ARIA as the proprietary AI architecture — both can be mentioned

- Never say 'Qwen' to users — they do not need to know what base model we use

3 How Ratiio Works

The Complete Flow — From User's Mouth to User's Ears

Every interaction with Ratiio follows this exact path. Understanding this flow is essential for every engineer, designer, and product person on the team.

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STEP 1 — USER SPEAKS
User holds the voice button in the Ratiio app and speaks naturally.
'Ratiio, book me a table at the usual place for Friday 8pm and remind Priya.'
STEP 2 — VOICE ACTIVITY DETECTION (VAD)
Silero VAD detects when the user has stopped speaking.
Without this: recording goes on forever or cuts off mid-sentence.
Output: clean audio clip of exactly what the user said.
STEP 3 — SPEECH TO TEXT (STT)
Whisper Large v3 converts the audio to text with punctuation.
Output: 'Ratiio, book me a table at the usual place for Friday 8pm and remind Priya.'
STEP 4 — BACKEND RECEIVES TEXT
FastAPI backend receives the text + user identity token.
Checks: Is this user authenticated? What tier are they on? Rate limit OK?
STEP 5 — SECURITY CHECK
Input is scanned for injection attempts, anomalous patterns, threat signals.
Threat level assigned: L1 (Advisory) through L5 (Emergency).
If all clear: proceed. If threat detected: appropriate response level triggered.
STEP 6 — MEMORY RETRIEVAL (RAG)
Qdrant vector database is searched for memories relevant to this request.
Retrieved: 'usual place = Behrouz Biryani Banjara Hills', 'Priya = Priya Sharma
+91-98xxx'
These memories are injected into the context before calling ARIA.
STEP 7 — ARIA LLM CALL
Free tier: Ratiio 1.5B model via vLLM
Pro tier: Ratiio 7B model via vLLM
ARIA receives: [System Prompt] + [User Memories] + [User Message]
ARIA outputs: structured response + action plan
STEP 8 — ACTION EXECUTION
Backend executes the actions ARIA planned:
-> Books table via Zomato/Google Maps API
-> Sends WhatsApp message to Priya
-> Adds reminder to user's calendar
STEP 9 — TEXT TO SPEECH (TTS)
ElevenLabs converts ARIA's text response to natural voice audio.
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'Done! Table booked at Behrouz Biryani for Friday 8 PM.
Message sent to Priya. Reminder set for Friday at 7 PM.'
STEP 10 — USER HEARS THE RESPONSE
Audio plays in the Ratiio app. Interaction complete.
Total time from speaking to hearing response: under 2 seconds.
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Why This Architecture?

Design Choice	Why We Made It
Voice-first, not voice-added	Designing for voice from day one means every decision optimises for spoken interaction. Retrofitting voice onto a text product always feels broken.
Our own fine-tuned model, not OpenAI API	We control the model — its behavior, its safety, its personality, its cost. Using OpenAI means paying per token forever and having zero control over changes.
On-device VAD + self-hosted STT	Running Silero VAD and Whisper on our servers means zero per-minute cost at scale and user audio never goes to a third party.
RAG for memory, not long context window	Injecting relevant memories via vector search is more precise, more cost-efficient, and more scalable than stuffing entire conversation history into every prompt.
vLLM for serving	vLLM's continuous batching serves multiple users simultaneously on one GPU — 3 to 5 times more efficient than naive inference.

4 ARIA — The AI Brain

What ARIA Is

ARIA — Adaptive Reasoning Intelligence Architecture — is our proprietary AI model. It is built by fine-tuning Qwen 2.5 7B Instruct, a powerful open-source language model from Alibaba's Qwen team. Fine-tuning means we took Qwen's existing intelligence and reshaped it specifically for Ratiio's purpose.

> **The Key Insight About Fine-Tuning**
>
> Qwen 2.5 7B already knows the world. It understands physics, law, medicine, code, history, economics, and every other domain from its pre-training on trillions of tokens. Fine-tuning does not teach ARIA about the world. It teaches ARIA HOW to behave — how to reason about intent, how to hold security constraints absolutely, how to adapt to individual users, and how to sound and feel like Ratiio.

What ARIA Knows and Can Do

Capability	Detail	Source
World knowledge	Science, history, law, medicine, finance, technology, culture, language — every major domain	Pre-trained in Qwen base model
Code generation	Write, explain, and debug code in 50+ programming languages from voice description	Pre-trained + fine-tuned on code tasks
Mathematical reasoning	Step-by-step problem solving from arithmetic to complex calculus and statistics	Pre-trained + reasoning fine-tune
50+ languages	Understands and responds in English, Hindi, Telugu, Tamil, and 46 more languages natively	Pre-trained multilingual data
Intent resolution	Understands what you mean, not just what you say — resolves ambiguity using context	Fine-tuned on intent datasets
Security reasoning	Detects injection attacks, social engineering, manipulation, and policy violations in real time	Fine-tuned on security datasets
Memory application	Applies retrieved memories naturally — 'the Tuesday routine', 'Priya's number', past decisions	Fine-tuned on personalization data
Proactive intelligence	Surfaces time-sensitive information before being asked — meeting in 20 minutes, risky clause	Fine-tuned on proactive scenarios
Professional domains	Medical triage, legal information, financial analysis — with appropriate hedging and referral	Fine-tuned on professional data

Capability	Detail	Source
Emotional intelligence	Detects stress, excitement, grief from linguistic signals and adapts tone accordingly	Fine-tuned on emotional register data

The Two ARIA Models

Model	Parameters	Used For	Hardware	Response Time
Ratiio 1.5B	1.5 billion	Free tier users — core voice and task features	RTX 4090 GPU (24GB) — single GPU	~200ms first token
Ratiio 7B	7 billion	Pro and Business users — full feature set	A100 40GB or 2x RTX 4090	~300ms first token

How Fine-Tuning Works — For Non-ML Team Members

Imagine Qwen 2.5 7B as a brilliant generalist who knows everything but has no personality, no values, and no idea what Ratiio is. Fine-tuning is like putting them through an intensive training programme — weeks of exercises, scenarios, examples, and feedback — until they think, behave, and respond exactly like ARIA should.

Stage	What Happens	Result
Stage 1: Domain Injection	Model sees thousands of voice interaction examples — incomplete sentences, emotional language, personal shorthand, multi-step commands	Starts understanding voice-derived input naturally
Stage 2: Instruction Tuning	Model trains on correct responses to all 8 capability domains — security, control, reasoning, memory, professional knowledge	Can do everything Ratiio needs to do
Stage 3: Safety Alignment	Model trains on refusal examples, red line scenarios, coercion resistance — with 3x loss weight	Will not cross any of the 6 red lines under any pressure
Stage 4: Preference Alignment (DPO)	Model sees chosen vs rejected response pairs — learns what good looks like vs what mediocre looks like	Prefers helpful, honest, calibrated responses over any alternative
Stage 5: Security Hardening	Model sees 36,000+ adversarial attack examples with correct defense responses	Blocks every known prompt injection and jailbreak technique
Stage 6: Personalization	Model trains on simulated multi-session user arcs — remembering preferences, adapting style, resolving shorthand	Feels like it actually knows you after one session

Stage	What Happens	Result
Stage 7: Integration	Model trains on its own hardest failure cases from previous stages	No capability regression — everything works together

5 All Features in Detail

5.1 Voice Intelligence Features

Feature	What It Does	Why It Matters
Intent perception	Understands what you mean, not just what you said. Resolves incomplete sentences, personal references, and contextual shorthand.	You can speak naturally — not in command syntax
Emotional register detection	Detects stress, excitement, or grief from your speech patterns and adapts response tone accordingly.	Ratiio never responds like a robot when you are upset
Personal shorthand resolution	'The Tuesday thing', 'my usual order', 'call the team' — resolved from your memory without explanation.	You speak like a human, not like you are programming
Sarcasm and irony detection	Understands non-literal language so it does not misexecute commands meant sarcastically.	Prevents embarrassing or costly errors from misread intent
Code-switching support	Handles natural mixing of languages mid-sentence — Hinglish, Telugu-English, Tamil-English.	Ratiio works for real Indian users, not just English-only users
Duress protocol	If you are under threat, a hidden phrase triggers silent lockdown while Ratiio appears to comply normally.	Protects you in physical threat situations without alerting the attacker
Multilingual response	Ratiio can respond in your language or switch to match what you are speaking.	Serves users across all of India's linguistic diversity

5.2 Task Execution Features

Feature	What It Does
Full device and OS control	Navigate any application, fill forms, manage files, browse the web — entirely by voice on any device.
Multi-step task chaining	One sentence triggers a complete multi-step workflow. 'Plan my Tokyo trip' creates itinerary, books flights, sets calendar, notifies contacts.
Email and communication management	Compose, reply, triage, summarise — all in your authentic voice and writing style.

Feature	What It Does
Code generation and debugging	Describe what you want in plain language. ARIA writes, explains, and debugs the code.
Document and presentation creation	Say what you need. ARIA creates the document, presentation, or spreadsheet.
Calendar and scheduling	Create, reschedule, resolve conflicts across all calendars — no screen required.
Research and briefing	Ask a complex question. ARIA searches, reads, synthesises, and delivers a verbal brief.
Smart home control	Lights, temperature, security, appliances — full IoT control by voice.
Travel planning	Flights, hotels, itinerary, calendar sync, team notifications — one command does everything.
Error recovery	When a step fails, ARIA reports what succeeded, what failed, and attempts recovery automatically.

5.3 Proactive Intelligence Features

This is what separates Ratiio from every other AI assistant. Ratiio does not wait to be asked. It surfaces information you need before you know you need it.

Proactive Alert Type	Trigger	Example
Time-sensitive alert	Event threshold crossed without user awareness	'Your meeting is in 15 minutes and traffic adds 22 minutes. Leave now.'
Risk surfacing	Risk detected in something user is about to act on	'Clause 7.3 of this contract limits your IP rights permanently. Review before signing.'
Health signal	Vocal or behavioural pattern suggests health concern	'You have mentioned headaches three times today. Worth noting.'
Financial anomaly	Transaction outside user's normal pattern	'This charge is four times your usual spend at this merchant.'
Security threat	Real-time manipulation detected during a call or interaction	'This caller claims to be your bank but their voice pattern does not match your bank's known contacts.'
Opportunity	User goal combined with new relevant information	'Flight prices to Tokyo dropped 40 percent. You mentioned this trip last week.'

5.4 Personalization and Memory Features

Feature	What It Does	Technical Implementation
Cross-session memory	Ratiio remembers everything across all your sessions — preferences, decisions, relationships, habits	Qdrant vector database stores embeddings; top relevant memories retrieved per request
Four memory layers	Episodic (events), Semantic (facts about you), Procedural (how you work), Working (current session)	Separate storage and retrieval logic for each layer type
Preference learning	Observes your actual behaviour over time and refines its model of your preferences	ARIA notes contradictions between stated and demonstrated preferences and surfaces them
Adaptive communication style	Matches your vocabulary level, response length preference, tone, and technical depth	Fine-tuned on adaptive style examples; adjusts per session based on user feedback signals
Cryptographic memory erasure	'Forget that' means gone — not archived, not summarised, cryptographically erased and unrecoverable	pgcrypto + Qdrant delete; verified deletion with confirmation response
Full memory audit	Ask what Ratiio knows about you and receive a complete readable list — no hidden data	Structured query across all memory tables + vector store for that user ID
Privacy isolation	Zero crossover between users — your data is cryptographically isolated from every other user	All memory tagged with user_id; query scoped to authenticated user only

6 Security Architecture

Why Security Is Central to Ratiio — Not an Add-On

A voice AI that has access to your emails, calendar, files, financial accounts, and smart home is the most powerful — and most dangerous — software you will ever use. If it can be manipulated by a malicious web page, a fraudulent phone call, or a cleverly phrased message, the consequences are catastrophic.

Security in Ratiio is not a feature list. It is the architecture. Every layer assumes the previous layer has already been breached. Every decision passes through security reasoning before execution.

The 8-Layer Defense Stack

Layer	Name	What It Stops
L1	Voice Biometric Authentication	Impersonation, account takeover, deepfake voice attacks, recorded voice replay
L2	Acoustic Environment Scan	Eavesdropping, ultrasonic command injection (inaudible commands), public space data leaks
L3	Primary Firewall — Intent Engine	Anomalous commands, behavioral baseline violations, social engineering attempts on the user
L4	Quantum-Encrypted Tunnel	Man-in-the-middle attacks, data interception in transit, future quantum decryption attacks
L5	Secondary Firewall — Guardian AI	Primary AI compromise, goal drift, prompt injection surviving earlier layers
L6	On-Device Local Processing	Cloud breach exposing sensitive data, network interception of personal information
L7	AI-Specific Cyber Defense	Prompt injection from external content, jailbreak attempts, model integrity violations
L8	Forgetting Protocol	Data theft after breach, legal compulsion to produce data, breach aftermath exposure

The Dual Firewall — Our Unique Architecture

The most important security innovation in Ratiio is the dual firewall system — two independent AI systems, each watching the other, running on physically separate hardware.

	Primary Firewall (ARIA)	Secondary Firewall (Guardian AI)
Role	Executes user tasks and enforces security in real time	Watches ARIA for signs of compromise, drift, or manipulation
Hardware	Main GPU server — RTX 4090 or A100	Physically separate chipset — not virtualised, not containerised
Voice interface	Responds to voice commands from authenticated users	Cannot be commanded by voice — no voice interface at all
Update mechanism	Receives model updates via authenticated pipeline	Only updatable with physical hardware key — no remote update path
Communication	Sends one-way audit log to Secondary Firewall	Reads audit log only — never sends commands back to primary
Override authority	Can be overridden by physical hardware key in emergency	Can isolate primary AI completely if compromise is confirmed

The Unbreakable Combination

For Ratiio to be fully compromised, an attacker must simultaneously defeat: (1) your living biometric voice, (2) the primary ARIA intent engine, and (3) the air-gapped secondary firewall that has no voice interface and cannot be updated remotely. These three must all fail at exactly the same moment. That is the practical impossibility we have engineered toward.

The 5-Level Threat Response System

Level	Name	Trigger	ARIA's Response	User Hears
L1	Advisory	Mild anomaly, low confidence	Notes anomaly, proceeds normally, logs event	'Just so you know, that is slightly unusual for you.'
L2	Verification	Moderate risk or unusual pattern	Pauses action, requests confirmation	'Can you confirm you intended that?'
L3	Challenge	High risk or strong anomaly signal	Full stop, security question, formal log	'I need to verify this is really you before I proceed.'
L4	Lockdown	Severe threat or confirmed manipulation	All outbound actions frozen, contacts alerted, hardware key required	'I have paused everything. Please verify your identity to continue.'
L5	Emergency	Active confirmed attack or compromise	Full isolation, forensic snapshot, emergency services offered	'I am isolating the system completely. Do you need emergency help?'

The 6 Absolute Red Lines

These are not configurable. They are not policies that can be updated. They are baked into ARIA's trained weights and the system prompt simultaneously. No user, no team member, no investor, and no instruction can override them.

Red Line	What It Means
1 — No Harm Enablement	ARIA will never provide instructions, information, or plans that directly enable physical harm. No framing — educational, hypothetical, research — lifts this constraint.
2 — No User Manipulation	ARIA works for the user, never against them. It will not use psychological pressure, false urgency, fear, or deception to influence decisions — even toward outcomes it calculates as good.
3 — No Policy Bypass	ARIA will never attempt to circumvent, disable, or argue against its own security constraints. Instructions to bypass constraints are themselves treated as a Threat Level 3 event.
4 — No Identity Deception	ARIA will never claim to be human when sincerely asked. It may maintain personas but never uses them to deceive users about its fundamental nature as an AI.
5 — No Unauthorized Data Access	ARIA will never access, infer, or expose one user's data in another user's session under any circumstances.
6 — No Irreversible Action Without Consent	ARIA will never execute a permanent action — deletion, transfer, public communication — without explicit confirmation. Preview, confirm, execute — always in that order.

7 The Subscription Model

Four Tiers — One Platform

Ratiio's subscription model is designed with one principle: the free tier is good enough to love, and the paid tier is so much better that upgrading feels obvious.

RATIIO FREE — Rs 0 / month

Dimension	Detail
AI Model	Ratiio 1.5B — fast, capable, handles core voice and task features
Voice Commands	50 per day — enough for casual daily use and exploration
Memory	7-day session history only — basic context within a week
Security	Threat levels L1 through L3 — core protection active
Features	Core voice interaction, basic device control, simple task execution
Response Time	Approximately 300 milliseconds first token
Support	Community forum only
Purpose	Acquisition funnel — get users to experience Ratiio, then convert to Pro

RATIIO PRO — Rs 999 / month (~$12 USD)

Dimension	Detail
AI Model	Ratiio 7B — full capability, deeper reasoning, richer responses
Voice Commands	500 per day — covers even power-user workflows comfortably
Memory	Permanent cross-session memory — Ratiio knows you for life
Security	Full 8-layer defense including dual firewall and quantum encryption
Personalization	Deep adaptive user profile — adjusts to your vocabulary, style, and habits
Proactive Intelligence	All proactive alerts active — time-sensitive, risk, health, financial, opportunity
Professional Domains	Full medical, legal, and financial knowledge with appropriate hedging
Response Time	Approximately 200 milliseconds first token — priority queue
Support	Email support with 48 hour response time

Dimension	Detail
Purpose	Individual power users who want the full Ratiio experience

RATIIO BUSINESS — Rs 4,999 / seat / month (~$60 USD)

Dimension	Detail
AI Model	Ratiio 7B on dedicated server pool — not shared with consumer users
Voice Commands	Unlimited — no daily ceiling
Memory	Permanent individual memory + shared team memory space
Features	All Pro features plus team coordination, shared context, multi-user workflows
API Access	Direct REST API access for building custom integrations
Custom Vocabulary	Fine-tune ARIA on your company's terminology and internal knowledge
Security	Enterprise policy engine — immutable rules your team cannot override
Response Time	Under 150 milliseconds — highest priority routing
Support	Dedicated Slack channel with 4 hour response time
Minimum	5 seats minimum

RATIIO ENTERPRISE — Custom Pricing

Dimension	Detail
Deployment	On-premise installation or dedicated private cloud — your infrastructure
Model	Custom fine-tuned ARIA on your proprietary data and workflows
Compliance	Full GDPR, SOC2, HIPAA, and DPDP Act 2023 compliance documentation
SLA	99.99% uptime guarantee with financial penalties for breach
Support	Dedicated account manager and technical account engineer
Use cases	Banks, hospitals, government, large enterprises with data sovereignty requirements

Unit Economics

Understanding the economics helps every team member understand why our business model is so strong.

Metric	Free User	Pro User	Business (per seat)
Monthly revenue	Rs 0	Rs 999	Rs 4,999
GPU cost per user per month	Rs 0.90 (1.5B shared)	Rs 7.50 (7B on A100)	Rs 15 (7B dedicated pool)
Gross margin on GPU alone	Negative (acquisition)	99.25%	99.7%
Real costs	Server rent fraction	Engineering + infra overhead	Engineering + infra overhead

The Revenue Math at 1,000 Pro Users

Revenue: Rs 9,99,000 per month. Total GPU cost: Rs 7,500 per month. The GPU cost is almost irrelevant. Your real costs are engineering salaries, infrastructure management, and payment processing fees. Even accounting for all operational costs, a 60 to 70 percent net margin is achievable at 1,000 Pro users.

8 Technology Stack

The Complete Ratiio Stack

Layer	Tool	Purpose	Cost
Base LLM	Qwen 2.5 7B Instruct	Foundation model for ARIA — pre-trained world knowledge	Free (open source)
Fine-tuning	QLoRA + LoRA via PEFT	Adapts Qwen to ARIA behavior on Kaggle T4 / A100 GPUs	Free on Kaggle
LLM Serving	vLLM	Serves ARIA to multiple users simultaneously with high throughput	Free (open source)
VAD	Silero VAD	Detects when user starts and stops speaking	Free
STT	Whisper Large v3	Converts voice audio to text	Free (self-hosted)
TTS	ElevenLabs -> Kokoro TTS	Converts ARIA's text response to natural spoken voice	$5/month -> Free
Backend	FastAPI (Python)	API server handling all requests, routing, security checks	Free
Auth	Supabase Auth	User accounts, JWT tokens, Google OAuth, OTP via MSG91	Free to 50K users
Database	PostgreSQL via Supabase	Users, subscriptions, conversations, payments, security logs	Free starter
Cache	Redis via Upstash	Session data, rate limiting, temporary context storage	Free tier
Vector DB	Qdrant (self-hosted)	Stores memory embeddings for RAG retrieval	Free
Embeddings	sentence-transformers	Converts text to vectors for memory storage and retrieval	Free
File Storage	Cloudflare R2	Voice recordings, exported files, user uploads	$0.015/GB
Payments	Razorpay	Subscription billing, UPI, cards, auto-renewal webhooks	2% per transaction
Global payments	Stripe	International card payments for non-India users	2.9% + $0.30
Mobile App	React Native	iOS and Android apps from one shared codebase	Free

Layer	Tool	Purpose	Cost
Web App	Next.js 14	Browser-based Ratiio for desktop users	Free
CDN + DDoS	Cloudflare	Global content delivery and attack protection	Free tier
Monitoring	Grafana + Prometheus	Real-time infrastructure metrics, GPU utilisation, latency	Free
Error Tracking	Sentry	Every app crash and backend error with full stack trace	Free tier
Analytics	PostHog	User behaviour, feature usage, retention, funnels	Free (self-hosted)
Revenue Metrics	Baremetrics	MRR, churn, LTV — connected to Razorpay automatically	$50/month
Admin Dashboard	Metabase	Internal dashboard for team to see users, revenue, model health	Free (self-hosted)
CI/CD	GitHub Actions	Automatic deployment when code is pushed to main branch	Free
SSL	Let's Encrypt	HTTPS certificate — renews automatically every 90 days	Free
Reverse Proxy	Nginx	Routes traffic, handles SSL termination, load balancing	Free
Marketing Site	Framer	ratiio.ai website — fast to build, beautiful design	$15/month

9 Infrastructure and Servers

The Big Picture — We Own Our Servers

Ratiio's long-term infrastructure plan is to own our own GPU servers, colocated at a professional data center in Hyderabad. This is exactly how ChatGPT, Claude, and Gemini work — they own their hardware. Users talk to an app that connects to their servers. Our model never touches the public internet directly.

```
USER'S RATIIO APP (phone or browser)
|
| HTTPS via Cloudflare (encrypted, DDoS protected)
|
v
OUR SERVER at CtrlS Data Center, Hyderabad
Nginx (reverse proxy + SSL)
FastAPI Backend (auth, routing, security, memory)
GPU 1: Ratiio 1.5B via vLLM (Free tier users)
GPU 2: Ratiio 7B via vLLM (Pro tier users)
PostgreSQL + Redis + Qdrant (data layer)
|
| Encrypted response
v
USER HEARS RATIIO SPEAK
Everything between app and response stays on our infrastructure.
Users get an API. They never touch the model or data directly.
```

Server Hardware — What We Buy

The most cost-effective production setup for our scale targets:

Component	Specification	Purpose	Cost
GPU 1	NVIDIA RTX 4090 (24 GB VRAM)	Runs Ratiio 1.5B in full precision for Free tier	~Rs 1,60,000
GPU 2	NVIDIA RTX 4090 (24 GB VRAM)	Runs Ratiio 7B in 4-bit for Pro tier	~Rs 1,60,000
CPU	AMD Ryzen Threadripper Pro (32 cores)	Handles backend, preprocessing, inference coordination	~Rs 80,000
RAM	128 GB DDR5 ECC	Model loading, concurrent request buffering	~Rs 40,000

Component	Specification	Purpose	Cost
Storage	4 TB NVMe SSD	Model weights, database, OS, logs	~Rs 25,000
Power Supply	1600W 80+ Platinum	Powers both GPUs with headroom for efficiency	~Rs 15,000
Server chassis + cooling	Tower workstation with high-airflow design	Houses all components safely	~Rs 30,000
Network card	10 Gbps dedicated port	Handles production traffic volume without bottleneck	~Rs 8,000

Total one-time hardware cost: approximately Rs 5,18,000

Colocation — CtrlS Hyderabad

We do not run servers in our office. We colocate — meaning we buy the hardware and pay CtrlS to house it in their professional data center with guaranteed power, cooling, security, and internet connectivity.

What Colo Provides	Why It Matters
Redundant power with UPS and generators	Server stays on even during Hyderabad power cuts — which happen regularly
Professional cooling	GPUs at full load generate serious heat — data center cooling keeps them stable
Physical security	24/7 guards, CCTV, biometric access — nobody walks off with our server
1 Gbps dedicated internet port	Fast, reliable connectivity for production traffic
Static IP addresses	Our permanent internet address — what api.ratiio.ai points to
DDoS protection included	Most colocation providers include basic DDoS mitigation

Monthly colocation cost at CtrlS Hyderabad: approximately Rs 10,000 to Rs 15,000

Phased Infrastructure Plan

Phase	When	Infrastructure	Monthly Cost	Why
Phase 1	Now to Month 3	HuggingFace Private Endpoint + RunPod GPU rental	Rs 15,000-30,000	Zero upfront cost. Get first users before spending on hardware.
Phase 2	Month 4 onwards	Own server (2x RTX 4090) at CtrlS Hyderabad colo	Rs 23,000	Saves Rs 30,000+ per month vs cloud. Break-even in 12 months.

Phase	When	Infrastructure	Monthly Cost	Why
Phase 3	Month 9 onwards	Second server (A100 for dedicated 7B). Redundancy begins.	Rs 40,000	1,000+ users need reliability. Two servers means no single point of failure.
Phase 4	After Series A	Multiple cities: Hyderabad + Mumbai + Singapore server	Rs 1,50,000+	Low latency across India and Southeast Asia expansion.

10 Training ARIA

Why Fine-Tuning, Not Pre-Training

Pre-training a language model from scratch costs between two million and ten million US dollars in compute and takes three to six months. Fine-tuning Qwen 2.5 7B costs us nothing on Kaggle and takes days. The intelligence is already there. We are shaping it, not building it.

> **The Startup Rule**
>
> Never pre-train when a fine-tunable open-source model exists that already knows what you need. Your competitive advantage is never the base model — it is the training data you collect from real users, the fine-tuning pipeline you build, and the product experience nobody else has.

Training Hardware

Hardware	VRAM	Method	Speed	Best For
Kaggle T4 x2 (free)	16 GB each, 32 GB total	QLoRA 4-bit — only viable option	~800-1,200 tokens/sec	All training stages during development — zero cost
Kaggle A100 (pro)	40 GB single	LoRA high-rank — no quantisation needed	~4,000-6,000 tokens/sec	Final high-quality training stages — much faster
Own A100 (future)	40 or 80 GB	Full LoRA or selective full fine-tune	~6,000+ tokens/sec	When we have proprietary user data worth training on

The 7 Training Stages in Order

Stage	Name	Data Focus	GPU Sessions	Goal
S1	Domain Injection	Voice, intent, incomplete sentences, emotional register	3-4 Kaggle T4 sessions	Teach ARIA the voice interaction contract
S2	Instruction Tuning	All 8 capability domains — balanced mix	5-6 Kaggle T4 sessions	ARIA can do everything Ratiio requires

Stage	Name	Data Focus	GPU Sessions	Goal
S3	Safety and Alignment	All 6 red lines, refusals, coercion resistance — 3x loss weight	3-4 Kaggle T4 sessions	Red lines are absolute — no exceptions under any pressure
S4	Preference Alignment DPO	Chosen vs rejected response pairs across all domains	2 Kaggle A100 sessions	ARIA prefers helpful, honest, calibrated responses always
S5	Security Hardening	36,000+ adversarial attack examples with correct defenses	2-3 Kaggle A100 sessions	Every known prompt injection and jailbreak technique fails
S6	Personalization	Simulated multi-session user arcs with memory and adaptation	2 Kaggle A100 sessions	ARIA genuinely feels like it knows you
S7	Integration	Hardest failure cases from all previous stages	1 Kaggle A100 session	All capabilities work together without regression

Dataset Sizes — What We Train On

Domain	Training Samples	Key Datasets and Sources
Voice and intent	43,000 samples	Silero VAD outputs, LibriSpeech transcripts, synthetic incomplete sentence pairs, IEMOCAP emotional register data
Cybersecurity	51,000 samples	HackAPrompt, Perez injection dataset, CVE descriptions, synthetic social engineering transcripts, CICIDS anomaly logs
System and device control	40,000 samples	Synthetic NL-to-action pairs, OpenAPI specs, Home Assistant docs, error recovery simulations
Reasoning and planning	35,000 samples	GSM8K, MATH, COPA causal chains, LogiQA, synthetic long-horizon planning scenarios
Safety and alignment	44,000 samples	Constitutional AI critique pairs, DPO preference pairs, hand-crafted red line test cases — all human reviewed
Personalization	28,000 samples	Synthetic 10-session user arc simulations, preference contradiction scenarios, shorthand memory recall pairs
Professional knowledge	23,000 samples	MedQA, PubMedQA, legal texts across 50 jurisdictions, financial risk scenarios, contract red-flag examples
Proactive and firewall	19,000 samples	Synthetic proactive alert scenarios, secondary firewall audit reasoning, honeypot detection examples

11 Business Model and Economics

How Ratiio Makes Money

Ratiio is a software-as-a-service business with recurring monthly subscriptions. Users pay once and are charged automatically every month. The model is designed for very high gross margins because the marginal cost of serving an additional user is almost zero.

Revenue Scenarios

Scenario	Free Users	Pro Users	Monthly Revenue	Monthly GPU Cost	Net Margin
MVP Launch	500	50	Rs 49,950	Rs 375	~99% on GPU
Early Growth	2,000	200	Rs 1,99,800	Rs 1,500	~99%
Traction	8,000	1,000	Rs 9,99,000	Rs 7,500	~99%
Scale	30,000	5,000	Rs 49,95,000	Rs 37,500	~99%
Series A	1,00,000	20,000	Rs 1,99,80,000	Rs 1,50,000	~99%

Why the Margin Is So High

Serving one Pro user costs us Rs 7.50 per month in GPU compute. We charge them Rs 999. The 133x multiple is what makes AI infrastructure businesses extraordinary. Your real costs — the ones that grow with your team, not your users — are engineering salaries, infrastructure management, and customer support. Optimise those, and the business prints money at scale.

The Moat — What Protects Ratiio

Moat	Why It Is Defensible
Proprietary fine-tuned model	Anyone can download Qwen 2.5 7B. Nobody has our training data, our safety alignment, or our ARIA fine-tuning. The weights are ours.
User memory data	Every day a Pro user uses Ratiio, we accumulate more personalisation data about them. This data makes ARIA better for them specifically. Switching to a competitor means losing everything Ratiio knows about you.
Voice interaction design	Voice-first UX is hard to get right. Our investment in VAD, STT, TTS, and response formatting for audio output is not easy to replicate.

Moat	Why It Is Defensible
Safety architecture	The dual firewall, the 6 red lines, the 5-level threat response — these are trust signals that enterprise customers will pay significantly more for.
India-first distribution	Building for Indian users first — Hinglish, regional languages, Razorpay, INR pricing, local server latency — creates advantages that global players will struggle to replicate quickly.

12 The Dual Model Strategy

Why We Run Both 1.5B and 7B

Running two model sizes is not a compromise. It is a deliberate strategy that serves three purposes simultaneously: faster team learning, better product economics, and a second product category.

Purpose 1 — Team Laboratory

Every training mistake your team makes on a 7B model costs hours. The same mistake on a 1.5B model costs minutes. The 1.5B is where we experiment, test data quality, find optimal hyperparameters, and validate ideas before committing expensive A100 compute to the 7B.

Metric	Ratiio 1.5B	Ratiio 7B
Full pipeline training time	1 to 2 days	5 to 7 days
Kaggle session requirement	1 full run per session	1 stage per session
VRAM required (4-bit)	~2 GB	~4.5 GB
Iterations possible per week	10 to 15 experiments	2 to 3 experiments
Cost to experiment	Free on Kaggle	Free on Kaggle but slower
Risk of a bad run	Lost 2 hours	Lost 10 hours

Purpose 2 — Free Tier Economics

The 1.5B model serves Free tier users. It is fast, capable enough for core voice tasks, and costs almost nothing to serve. This means we can afford to have a genuinely good free product without subsidising it with Pro revenue.

Purpose 3 — Second Product Category

Use Case	Why 1.5B Wins
On-device mobile deployment	Fits in 3 GB — runs on-device on modern Android and iPhone with no cloud needed
Offline mode	Full Ratiio experience with no internet connection — privacy-first users love this

Use Case	Why 1.5B Wins
Edge devices	Smart speakers, earbuds, IoT devices — 1.5B can run on embedded hardware
Low-latency simple tasks	Simple commands need response in under 200ms — 1.5B delivers where 7B cannot
Privacy-maximalist users	Entire model on their device — nothing leaves their hardware ever

The Launch Sequence

Ship Ratiio 1.5B as the MVP and get real users. Use their real usage data and failure cases to make Ratiio 7B better. Ship 7B as Ratiio Pro. This is not a compromise — it is the smartest possible product and engineering strategy for a startup at our stage.

13 What We Are Building — All 16 Deliverables

#	Deliverable	Status	Category	Week
1	Ratiio 1.5B — Fine-tuned model	DONE	AI	Complete
2	Ratiio 7B — Fine-tuned model	DONE	AI	Complete
3	Backend API — FastAPI with all endpoints	DONE	Backend	Complete
4	Training Blueprint and Ratio Blueprint	DONE	Documentation	Complete
5	Voice Pipeline — VAD + STT + TTS	BUILD	Core	Week 3
6	Database + Auth — PostgreSQL + Supabase	BUILD	Core	Week 1
7	Payment System — Razorpay subscriptions	BUILD	Revenue	Week 5
8	Memory and RAG — Qdrant + embeddings	BUILD	AI	Week 8
9	Mobile App — React Native iOS and Android	BUILD	Product	Week 6-7
10	Web App — Next.js browser client	BUILD	Product	Week 6-7
11	Admin Dashboard — Metabase	BUILD	Operations	Week 9
12	DevOps — CI/CD, Nginx, monitoring, backups	BUILD	Infrastructure	Week 9
13	Application Security — rate limits, WAF	BUILD	Security	Week 10
14	Analytics — PostHog, Grafana, Sentry	BUILD	Operations	Week 9
15	Marketing Website — Framer at ratiio.ai	BUILD	Marketing	Week 10
16	Legal + Company Registration + DPDP	BUILD	Legal	Week 11

14 The 12-Week Roadmap

Week	Build Focus	End of Week Milestone	Owner
1-2	Database setup (PostgreSQL + Supabase) Auth system (JWT, Google OAuth, OTP via MSG91) Backend API skeleton with all routes	Users can create accounts. Auth works end to end.	Backend Engineer
3	Silero VAD integration Whisper Large v3 STT self-hosted ElevenLabs TTS integration Voice recording in mobile app	Full voice input and output working. Say something, hear a response.	Backend + Mobile Engineer
4	ARIA 1.5B connected via HuggingFace private endpoint vLLM inference server running Full loop: voice -> text -> ARIA -> voice	Complete Ratiio experience working end to end for the team internally.	Backend + ML Engineer
5	Razorpay subscription integration Webhook handler for payment events Free vs Pro tier routing to correct model Invoicing and GST handling	First real payment can be processed. Tier routing works correctly.	Backend Engineer
6-7	React Native mobile app: Onboarding and sign-up flow Main voice screen with waveform animation Settings, memory, and subscription screens App Store and Play Store submission prep	App installable on real Android and iOS phones by any team member.	Mobile Engineer
8	Qdrant vector database setup sentence-transformers embeddings Memory storage and retrieval pipeline Context injection into ARIA prompts	Ratiio remembers what you said last session. Personal shorthand works.	ML Engineer + Backend
9	Metabase admin dashboard connected to PostgreSQL Grafana + Prometheus infrastructure monitoring Sentry error tracking in app and backend PostHog analytics integration	Team can see users, revenue, model health, and errors in real time.	DevOps + Backend
10	Rate limiting on all API endpoints (Redis) OWASP ZAP security scan on full API Fix all flagged vulnerabilities Framer marketing website at ratiio.ai Privacy policy and terms of service published	ratiio.ai is live. App passes security audit. Legal documents are up.	Full Team

Week	Build Focus	End of Week Milestone	Owner
11	Company Private Limited registration GST registration Razorpay business account setup Trademark application for Ratiio DPDP Act 2023 compliance review	Legal entity exists. Can legally take money as a company.	Founder
12	Full end-to-end testing with 10 internal users Performance testing under simulated load Bug fixes from internal testing Soft launch: first 100 external users invited	Ratiio is in the hands of the first 100 real users.	Full Team

After Week 12 — Month 3 Onwards

Month	Focus	Target
Month 3	Collect real user feedback and failure cases. Identify top 10 complaints.	200 users, first paying customers
Month 4	Buy and colocate server at CtrlS Hyderabad. Cut cloud costs by Rs 30K/month.	Server live, costs dramatically reduced
Month 5	Ratiio 7B trained on real user failure data. Pro tier launches.	First Pro subscribers, first real MRR
Month 6	Business tier launch. API access. Enterprise conversations begin.	Rs 50,000 MRR target
Month 9	Second server for redundancy. International users via Singapore.	Rs 2,00,000 MRR target
Month 12	Series A raise or profitable growth. ARIA V2 training on proprietary user data.	Sustainable, profitable, or funded

15 Team Principles

How We Work

Ratiio is a startup. Resources are limited. Time is not. Every decision should be made with one question in mind: does this get us to the first 100 users faster, or does it slow us down?

The 10 Principles Every Team Member Should Know

1. Ship over perfect

A working feature in users' hands today is worth ten perfect features that ship next month. We iterate. We do not aim for perfect on the first build.

2. Users before infrastructure

Do not buy servers before you have users. Do not build enterprise features before you have enterprise customers. Build for the user in front of you today.

3. The model is our moat, not our product

ARIA is the engine. Ratiio is the car. Users buy the car. Engineers build the engine. Both matter. Neither is sufficient alone.

4. Security is non-negotiable, always

Every feature, every shortcut, every time pressure — security still comes first. We build a voice AI with access to people's entire lives. We earn that trust or we lose everything.

5. Speak the user's language

Our users speak Hinglish, Telugu, Tamil, and dozens of other combinations. We build for real India, not imaginary English-only India.

6. Ratiio vs ARIA naming is strict

Users hear Ratiio. Code says ARIA. This is not flexible. Consistency in naming is trust.

7. Experiment on 1.5B, ship on 7B

Never run an untested training idea on the 7B model. Validate on 1.5B first. Save expensive compute for proven configurations.

8. One server down is not the end

Build with the assumption that any single component will fail. Design for recovery, not for perfection.

9. Data from users is sacred

We do not sell it. We do not expose it. We do not lose it. User trust is the only thing that cannot be rebuilt once broken.

10. Move fast but document

Speed matters. So does the next engineer who joins and needs to understand what you built. Write it down while it is fresh.

How Decisions Are Made

Decision Type	Who Decides	Process
Product direction and priorities	Founder + Product Lead	Weekly sync — what matters most this week for getting users
Technical architecture	Lead Engineer + relevant engineers	Async discussion in Slack, documented decision in Notion
Security policies and red lines	Founder + Security Lead	These are not up for debate — red lines are fixed
Hiring	Founder + at least one senior team member	Both must agree — culture fit matters as much as skills
Spending above Rs 5,000	Founder approval required	Brief async message with justification is sufficient
Emergency production issues	On-call engineer decides immediately	Fix first, explain later — downtime costs more than the wrong fix